



上海科技大学
ShanghaiTech University

EE115 Analog Circuits 模拟电路

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Outline



- Ideal Op Amp
- SEDTRA/SMITH book page 59-81

Ideal Op Amp

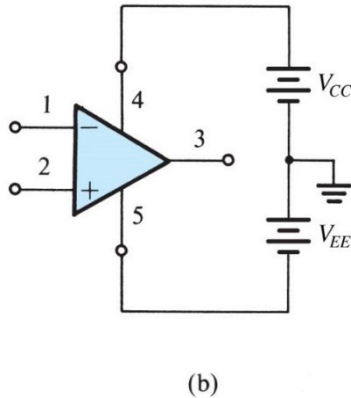


Figure 2.2 The op amp shown connected to dc power supplies.

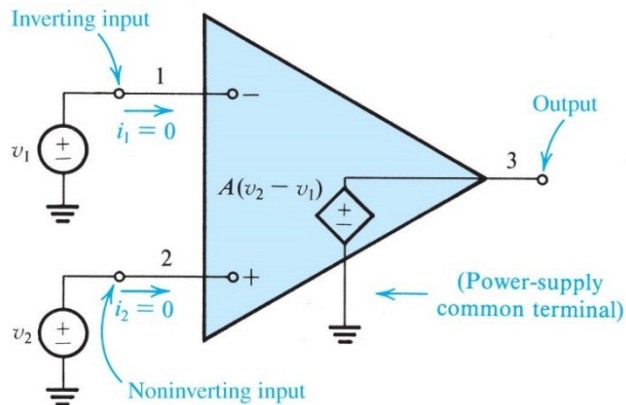


Figure 2.3 Equivalent circuit of the ideal op amp.

- Infinite input impedance
 - No current goes in
- Zero output impedance
- Infinite open-loop gain, A
 - How is the possible?
 - Use feedback to define circuit gain
- Infinite bandwidth
- Infinite common-mode rejection

Inverting Amplifier

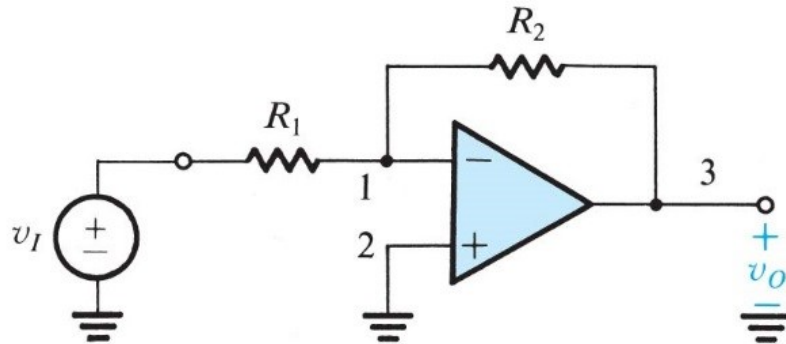
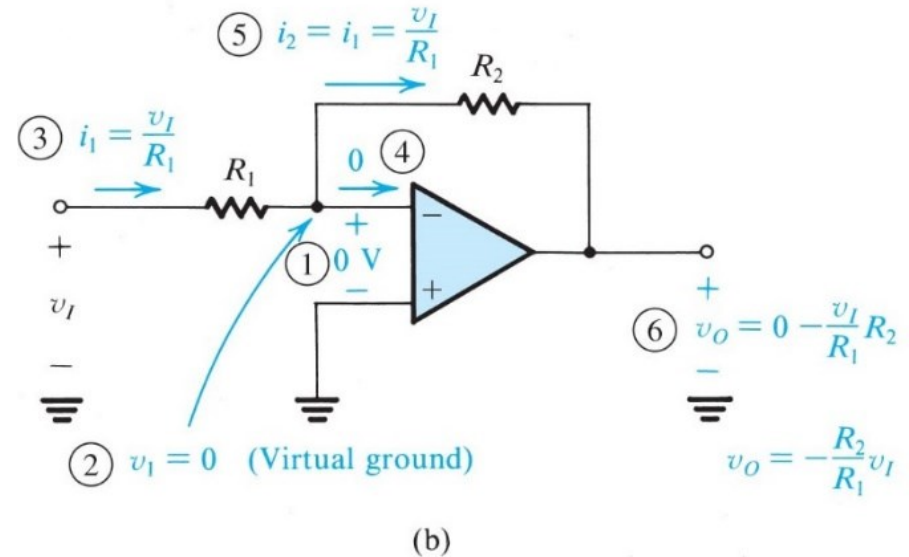
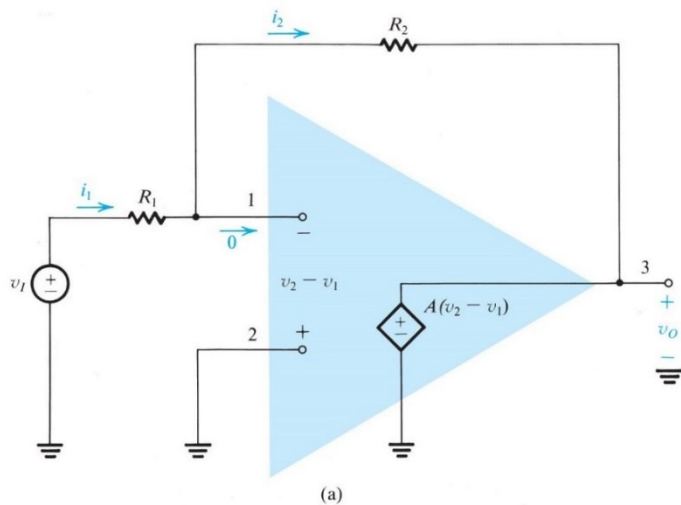


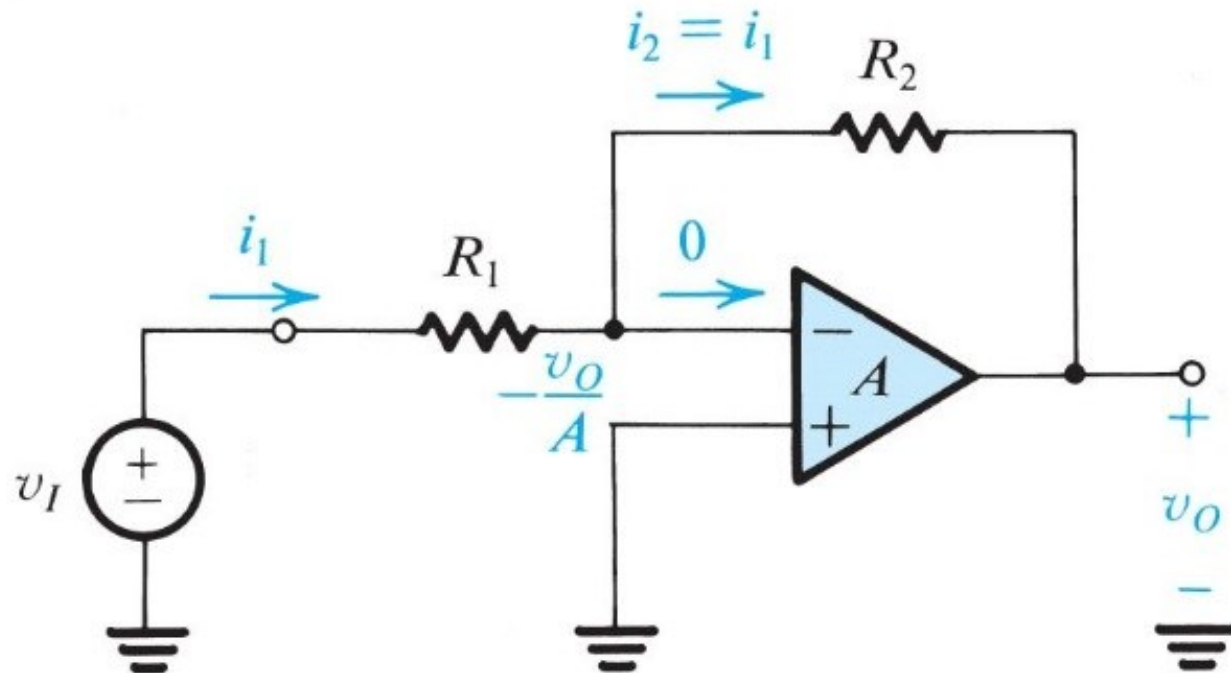
Figure 2.5 The inverting closed-loop configuration.



$$\frac{v_O}{v_I} = -\frac{R_2}{R_1}$$

Figure 2.6 Analysis of the inverting configuration. The circled numbers indicate the order of the analysis steps.

Inverting Amplifier with Finite Open-Loop Gain



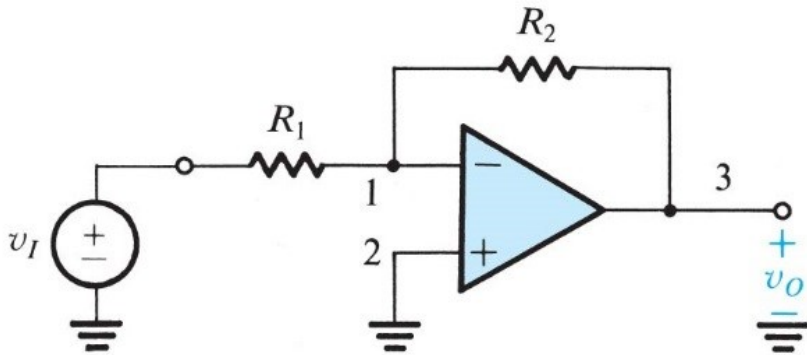
$$i_1 = \frac{v_I - (-v_O/A)}{R_1} = \frac{v_I + v_O/A}{R_1}$$

$$\begin{aligned} v_O &= -\frac{v_O}{A} - i_1 R_2 \\ &= -\frac{v_O}{A} - \left(\frac{v_I + v_O/A}{R_1} \right) R_2 \end{aligned}$$

Figure 2.7 Analysis of the inverting configuration taking into account the finite open-loop gain of the op amp.

$$G \equiv \frac{v_O}{v_I} = \frac{-R_2/R_1}{1 + (1 + R_2/R_1)/A}$$

Inverting Amplifier: Input and Output Resistances



Since $v_I = 0$

- The input resistance is simply $R_i = R_1$

- To find output resistance
 - 1. Short-circuit the source
 - 2. Apply a test source at the output, and find the impedance

Application of Inverting Amplifier: Weighted Summer

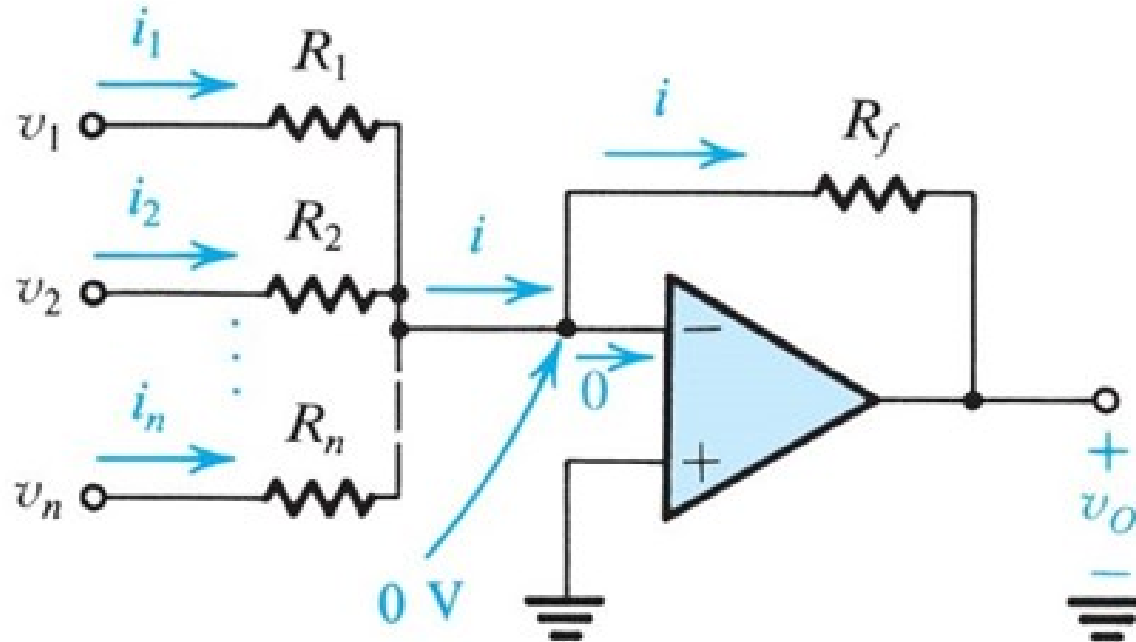


Figure 2.10 A weighted summer.

$$v_O = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \cdots + \frac{R_f}{R_n}v_n\right)$$

Non-Inverting Amplifier

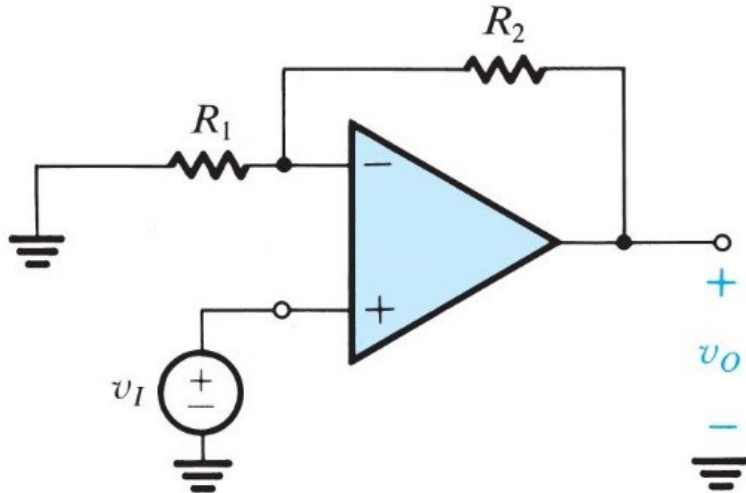


Figure 2.12 The noninverting configuration.

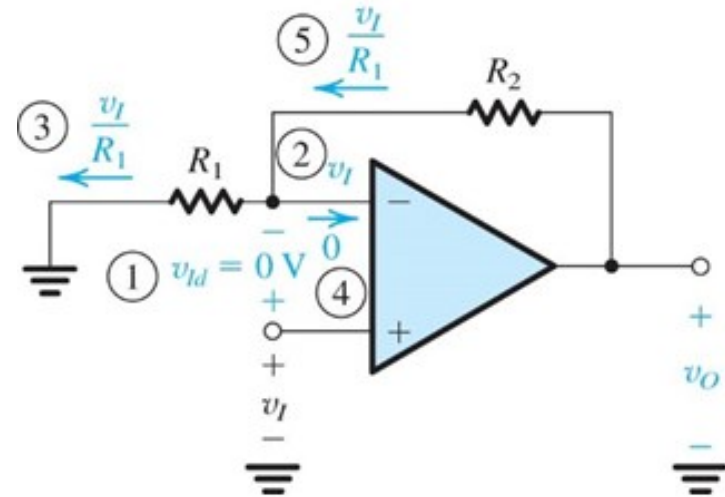


Figure 2.13 Analysis of the noninverting circuit. The sequence of the steps in the analysis is indicated by the circled numbers.

$$\frac{v_O}{v_I} = 1 + R_2/R_1$$

Non-Inverting Amplifier: Effect of Finite Open-Loop Gain

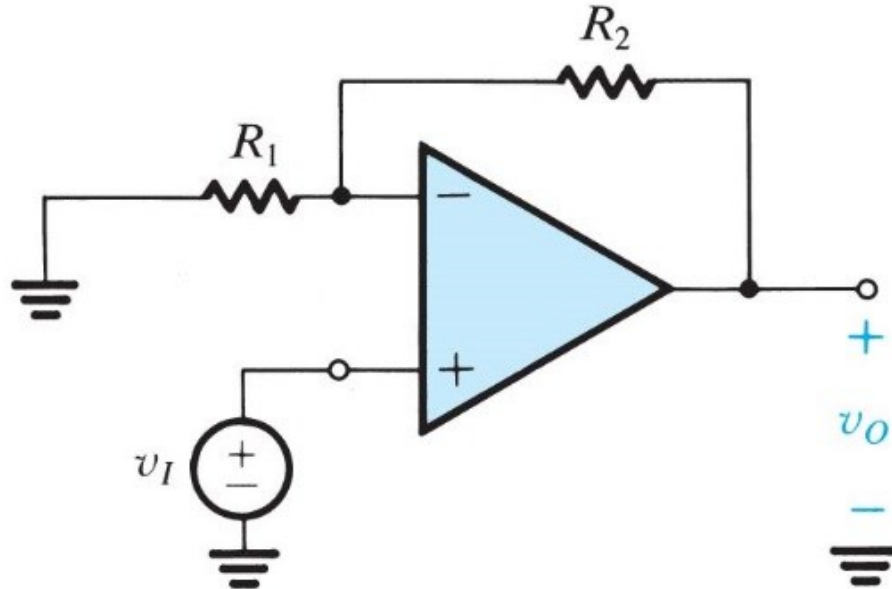


Figure 2.12 The noninverting configuration.

$$G \equiv \frac{v_O}{v_I} = \frac{1 + (R_2/R_1)}{1 + \frac{1 + (R_2/R_1)}{A}}$$

Non-Inverting Amplifier: Input and Output Resistances

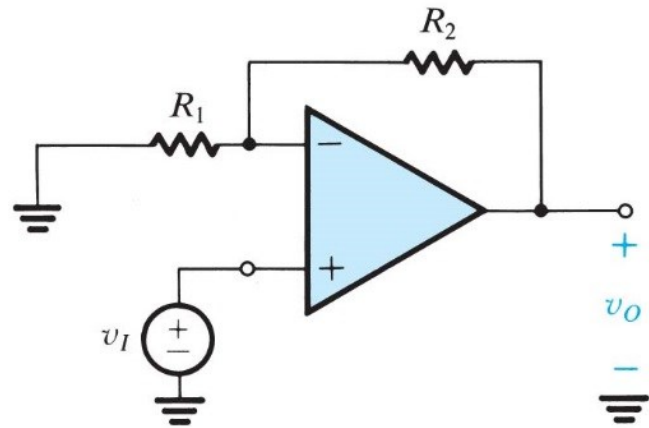


Figure 2.12 The noninverting configuration.

- For ideal Op-Amp
- No current flows into the amplifier
- $R_i = \infty$
- To find output resistance
 - Replace v_I with short circuit
- $R_o = 0$

Voltage Follower: Unity Gain Amplifier

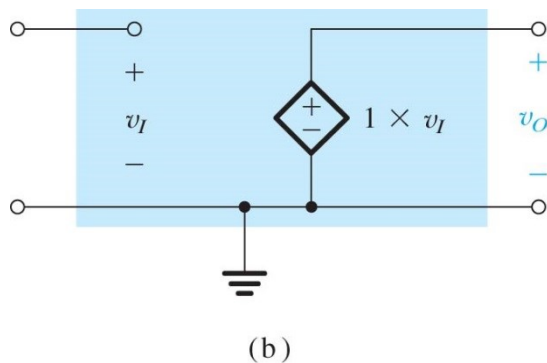
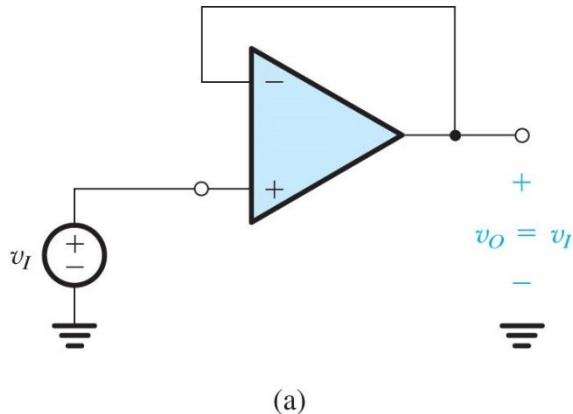


Figure 2.14 (a) The unity-gain buffer or follower amplifier. (b) Its equivalent circuit model.

- Unity voltage gain
 - But large power gain
- Purpose:
 - Provide impedance transformation:
 - A buffer stage that present infinite input impedance to the source, and zero output impedance to the load
 - Power amplifier
 - Boost drive current

Differential Amplifier

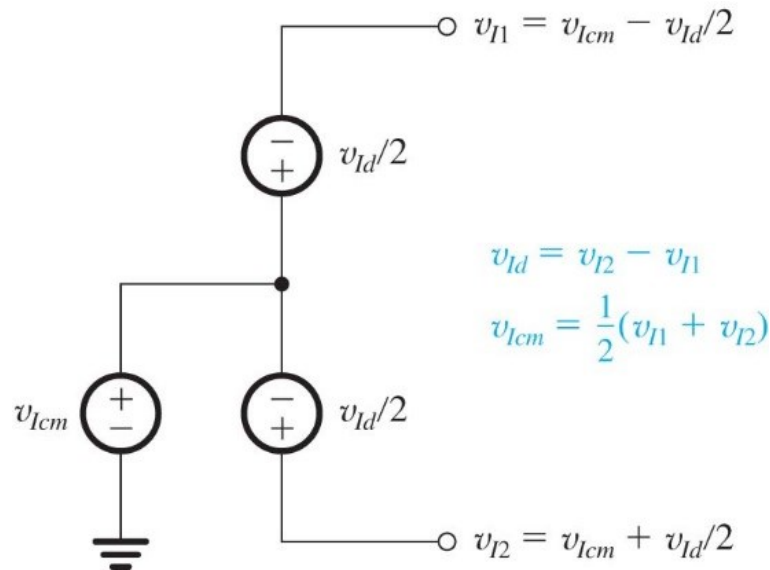


Figure 2.15 Representing the input signals to a differential amplifier in terms of their differential and common-mode components.

- A differential amplifier is one that **responds to the difference** between the two signals applied at its input and ideally **rejects** signals that are **common** to the two inputs
- $v_o = A_d v_{Id} + A_{cm} v_{Icm}$
- Common-mode rejection ratio (CMRR)
- $CMRR = 20\log(|A_d|/|A_{cm}|)$

Differential Amplifier: Differential gain

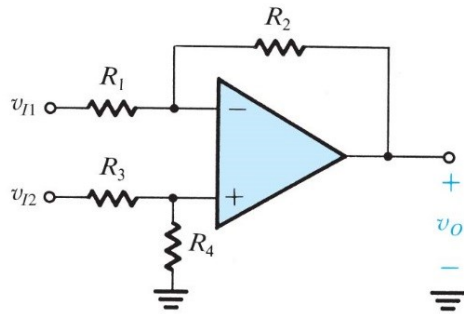


Figure 2.16 A difference amplifier.

- Apply superposition method

$$v_{O1} = -\frac{R_2}{R_1} v_{I1} \quad \text{Condition: } \frac{R_4}{R_3} = \frac{R_2}{R_1}$$

$$v_{O2} = v_{I2} \frac{R_4}{R_3 + R_4} \left(1 + \frac{R_2}{R_1} \right) = \frac{R_2}{R_1} v_{I2}$$

- Differential gain

$$v_O = \frac{R_2}{R_1} (v_{I2} - v_{I1}) = \frac{R_2}{R_1} v_{Id}$$

$$A_d = \frac{R_2}{R_1}$$

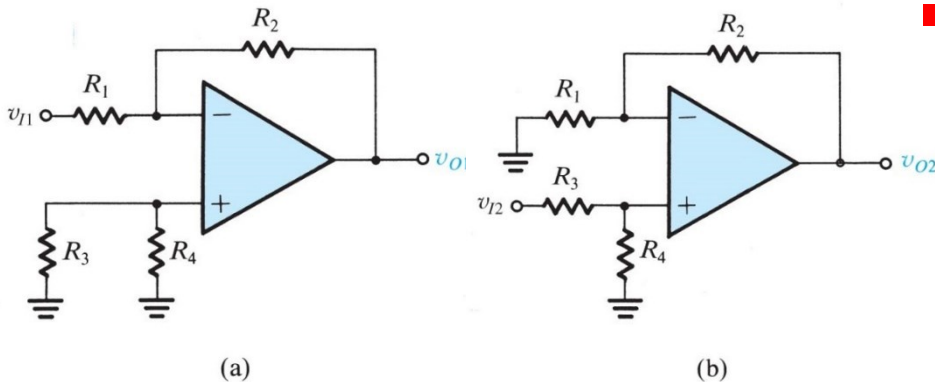
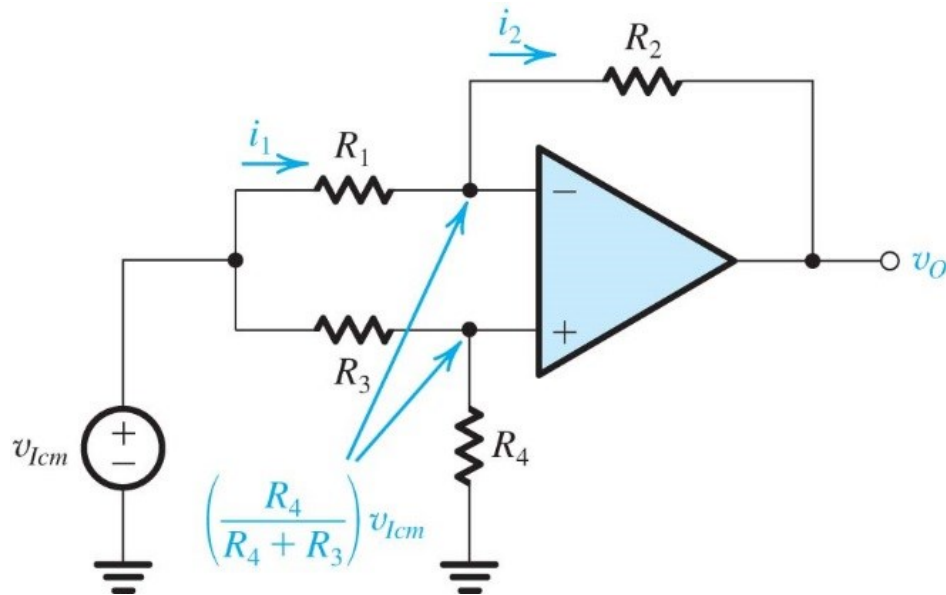


Figure 2.17 Application of superposition to the analysis of the circuit of Fig. 2.16.

Differential Amplifier: common mode gain



$$A_{cm} \equiv \frac{v_O}{v_{Icm}} = \left(\frac{R_4}{R_4 + R_3} \right) \left(1 - \frac{R_2 R_3}{R_1 R_4} \right)$$

$$\text{Condition: } \frac{R_4}{R_3} = \frac{R_2}{R_1}$$

$$A_{cm} = 0$$

Figure 2.18 Analysis of the difference amplifier to determine its common-mode gain

$$A_{cm} = v_o / v_{cm}$$

Differential Amplifier: input resistance

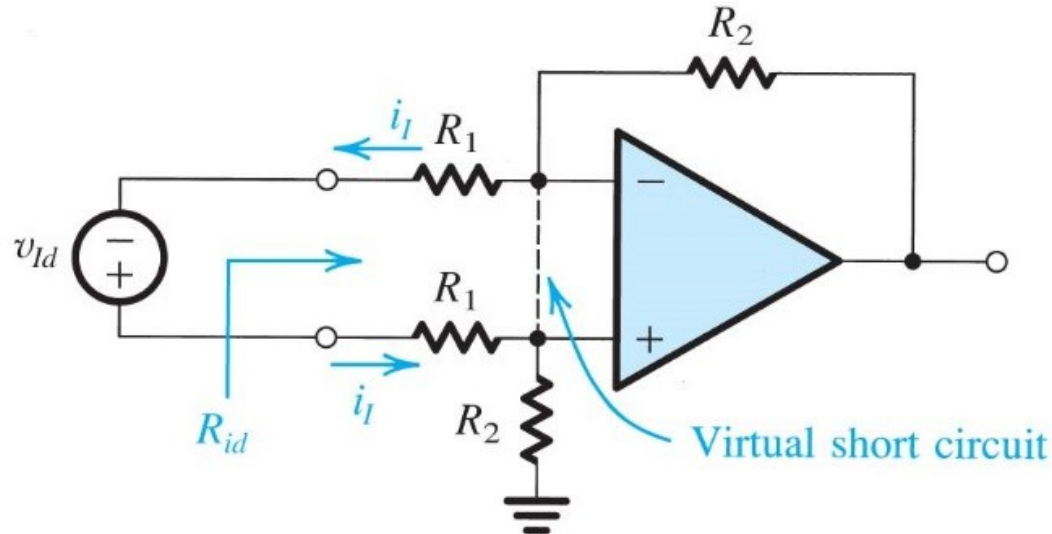
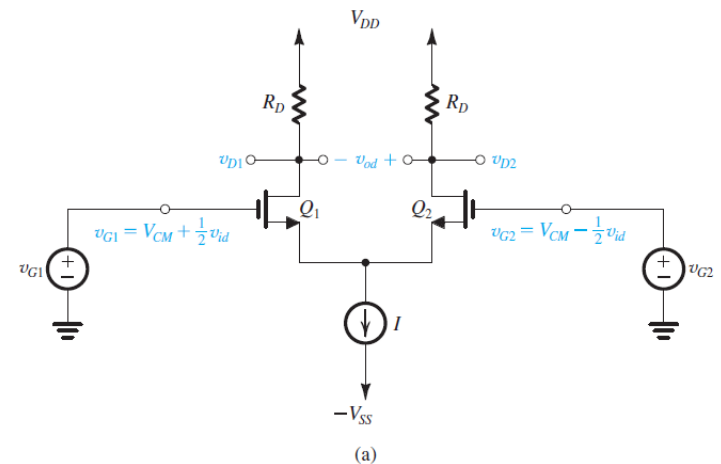


Figure 2.19 Finding the input resistance of the difference ampl
 $R_3 = R_1$ and $R_4 = R_2$.

$$R_{id} = 2R_1$$



Homework1-due March 18th 11p.m.



SEDRA/SMITH Book chapter 1 problem 1.19, 1.43,
1.68

Chapter 2: problem 2.8, 2.51, 2.60